Car Fleet, Scale & Protection (all games)

2026-09-04. The fleet + damage strategy that gates BOTH events (repair time/cost is the ops killer). Covers Yash's asks: TorqFall #3 (protect cars so bodywork survives), TorqSiege #2 (crash without damage), TorqSiege #3 + cross-cutting (which scale, which specific models, options). Model data grounded by a Sept-2026 India-market scan. ***Boundary:*** *this is the* ***game-design*** *fleet call (scale/model/protection). Final* ***procurement + landed cost*** *is owned by the parallel Sourcing thread — hand them this shortlist; prices below are retailer listings that move with stock/FX and must be confirmed at checkout.*

> **⚠ SUPERSEDED WHERE IT CONFLICTS (2026-09-06): the locked fleet decision is RENT, not buy.** The fleet is 12–16 rented stock cars per event from an RC club partner (they bring batteries, spares + pit crew — see [[TorqFall — What The Event Needs]] §1), and **nothing is bolted, screwed, glued or stripped on a rented car — strap-on/velcro/zip-tie only.** That kills three things below as written: the buy quantities (§1), body-stripping (Layer 1) and the per-car printed TPU cage (Layer 2). On the rented fleet the whole protection story is **zip-tied foam/noodle bump rings (Layer 2's fallback) + the speed cap (Layer 4)**; the siege ram is rented from the same club (₹2–4k/day uplift). Everything else here — scale logic, model shortlist, what-breaks analysis — stands as (a) the spec of what to ask the rental club for and (b) the buy plan for a **future owned fleet**. Fleet count for planning = **12–16** (the Needs-doc derivation: 6–8 live/heat + 4–6 charging + 2 spares); the 16–20 in [[Fleet & Scale Decision v1]] was the old two-zones-running-at-once frame and is superseded by the one-game-per-event lock.

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1. Scale decision

Two scales, one shared philosophy — reuse the fleet across both games for capital efficiency.

- **Workhorse fleet = 1/14 (or 1/16), speed-limited, brushed.** Small = safer in a pack, less crash energy, cheapest spares, most units. Serves TorqFall's pack heats AND TorqSiege's builder/light-attack roles. Fleet size **12–16 — rented per event** (see banner; "buy 12–16" applies only to a future owned fleet).
- **Siege rams = 1/10, brushless, throttle-capped. 1–2 — rented from the same club** (flag in the car deal, ₹2–4k/day uplift; buy only for a future owned fleet). A 1/10 at *low speed* hits far harder than any 1/14 — the mass/torque you need to reliably topple foam/cardboard. Used only in TorqSiege attack rounds (and to prove the collapse mechanic).

Why not one scale for everything: 1/16 can't guarantee a dramatic collapse (light + soft tyres = soft hit — the whole TorqSiege physics risk); 1/10 in an 8-car TorqFall pack is too much energy near a crowd. Split the roles.

2. Specific models (India-available, spares-backed — Sept 2026)

Model choice should be driven by **India-domestic spare-parts pipelines**, not price — the top models have dedicated India parts pages (Daddy Drones, Bharat Hobby, Hobby Central), so we're not AliExpress-dependent.

Workhorse fleet (pick one, standardise the whole fleet on it):

Model	Scale	Motor	~INR	Why
WLtoys 144001 ⭐	1/14	Brushed 550	~12,990	**Default.** Most parts-supported RC on earth (India *and* global); every failure part is pennies in bulk; ~60 km/h caps easily to ~30 for rental.
WLtoys 144018	1/14	Brushed	~9,990	Cheapest; **35 km/h stock = ideal rental speed with zero limiting effort.** Good mix-in.
MJX Hyper Go 16208 / 14208	1/16–1/14	Brushless	~14–17k	**Modern alt:** metal diffs, and **clipless body = seconds to swap a cracked shell mid-event** (big ops win for a crash fleet). Best India spares of the newer models. Needs app/TX throttle-limiting.

Siege ram (buy 1–2):

Model	Scale	~INR	Why
WLtoys 104019 ⭐	1/10	~19,990	**Best value ram** with India spares; enough mass to topple structures at capped speed.
MJX Hyper Go 10208 V2 / 10210	1/10	~30–40k	Max mass + torque if 104019 under-delivers in the collapse test; heavier hit, clipless body.

Avoid for a crash fleet: ZD Racing (thinner India spares), entry Arrma (import-only spares, expensive). Price flags: exact India RTR prices for MJX 16207/16208/14209/10208 weren't printed on retailer pages — models are in stock; budget the ranges and confirm at checkout.

3. Brushed vs brushless

- **Fleet = brushed.** Durability from restraint (35–45 km/h = less crash energy), trivial speed-limiting (TX throttle EPA, or just the brushed motor), cheap ₹200–400 motor swaps, tolerant of stalling against a wall. Brushless ESCs are the expensive impact-failure point.
- **Siege ram = brushless** (you *want* torque to shove structures) — run it throttle-capped via the MJX app / TX EPA.

4. The damage problem — how we crash cheap (both games)

What breaks, in order (all these models): (1) body shell / posts, (2) rear suspension arms + steering knuckles, (3) dogbones / drive cups, (4) diff gears / pinion, (5) wheels/tyres, (6) servo. Strategy attacks this in layers:

Layer 1 — remove what breaks *(owned-fleet only — you cannot strip a rented car's body; on rentals, the foam ring below takes the hits instead)*. Strip stock scale bodies of wings/mirrors/detail (they snap first and add nothing). Run **bare or a purpose-built protective shell**, not the fragile stock polycarbonate.

Layer 2 — soft armor cage (Yash's idea, validated). *On the RENTED fleet the "fallback" IS the primary — nothing prints onto or fixes to a rented car:* **EVA foam / pool-noodle bump rings zip-tied around the perimeter** (come off clean at handback) — near-zero cost, no printer needed; this is the line priced in [[TorqFall — What The Event Needs]] §1. *Owned-fleet upgrade:* 3D-print a **TPU** (flexible filament) **exoskeleton bumper-ring / roll-cage** around the chassis perimeter. TPU is exactly right — it's *elastic*: it absorbs and rebounds impacts instead of shattering like PLA/ABS. A full-perimeter TPU bump-ring protects arms/knuckles (the #2 breakage) by taking side-hits first. Print once per car — once the cars are ours.

Layer 3 — quick-swap consumables (repair = swap, not fix). Design the pit so the parts that *do* break are tool-free, bulk-stocked swaps. **Standardise the fleet on ONE model** so one spares bin covers every car. Clipless-body models (16208) swap a shell in seconds. Pre-buy per the pit list below.

Layer 4 — detune (already in plan). Speed cap (energy ∝ v^2) + soft rubber tyres already chosen — the cheapest damage reduction there is.

Layer 5 — TorqSiege-specific (controlled crashing). The ram takes the load head-on → give it a **sacrificial replaceable front ram plate/wedge** (PLA/plywood, cheap, expected to wear — **velcro-strap-mounted on the rented ram**, per [[TorqSiege — Build Phase & Weapons v1]] §5) and a low **wedge geometry that rides up into the wall and shoves, not nose-dives** (protects wheels/suspension — the vulnerable bits when hitting structure). **Box and protect the electronics** (ESC/RX/servo) behind the ram plate. The car is designed to *deliver* the hit and survive it; the wedge is the fuse.

5. The pit — pre-buy list (per event)

Owned-fleet frame. With the rented club deal, spares + pit crew come WITH the club (that's the point of hiring the club, not just the cars — [[TorqFall — What The Event Needs]] §1); this list doubles as what to ask the club to bring. Standardise on one workhorse model, then bulk-buy from one India shop:

- **2× spare body shells per car** (clipless if 16208).
- **Bulk suspension-arm + steering-knuckle set.**
- **Spare dogbones + diff cups.**
- **1–2 spare servos per 4 cars.**
- Spare wheels/tyres; a couple of spare ESCs (brushless ram).
- TPU bump-rings printed as spares; foam-ring fallback kit.
- Charging: spare battery sets per car for always-on rotation (see [[Event Ops Standards (all games)]] §5).

Next: (a) borrow/rent 1 workhorse + 1 ram to prototype the zip-tied foam ring + the strap-on siege ram plate and run the collapse test ([[TorqSiege — Collapse Test Protocol v1]]); (b) hand this shortlist to the Sourcing thread for landed cost on a *future owned fleet* (rent is locked for the event fleet — see banner); (c) finalise the speed-limiting method per model in rehearsal. Pairs with [[TorqSiege — Build Phase & Weapons v1]] and [[Event Ops Standards (all games)]].